USART Driver Analysis

The USART (Universal Synchronous/Asynchronous Receiver Transmitter) driver is used to enable serial communication between the STM32F446xx microcontroller and external devices such as PCs, sensors, Bluetooth modules, GPS modules, and other microcontrollers.

How the Driver Works

The driver is built around two main structures: USART_Config_t and USART_Handle_t.

The USART_Config_t structure stores communication settings such as baud rate, operating mode (Transmit, Receive, or Both), word length, parity control, stop bits, and hardware flow control.

The USART_Handle_t structure contains the USART peripheral address, configuration settings, transmit and receive buffers, data lengths, and communication status information.

When USART_Init() is called, the driver first enables the peripheral clock and then configures the USART registers according to the selected settings. The driver sets up:

- Communication mode (TX, RX, or TX/RX)
- Baud rate
- Word length (8-bit or 9-bit)
- Parity control
- Stop bits
- Hardware flow control (CTS/RTS)

After configuration, the peripheral can be enabled using USART_PeripheralControl().

How the Driver Hides Hardware Complexity

Normally, configuring USART requires direct access to several hardware registers such as CR1, CR2, CR3, BRR, SR, and DR. The programmer must also calculate baud rate values and manage interrupt settings manually.

This driver hides these low-level details by providing simple APIs. The user only needs to fill the configuration structure and call the required functions. The driver automatically performs register configuration, baud rate calculations, data transmission, reception, and interrupt management.

This makes application development easier and reduces programming errors.

APIs Offered by the Driver

Clock and Peripheral Control

- USART_PeriClockControl()
    - Enables or disables the clock for a USART peripheral.
- USART_PeripheralControl()
    - Enables or disables the USART peripheral.

Initialization Functions

- USART_Init()
    - Configures baud rate, word length, parity, stop bits, operating mode, and flow control.
- USART_DeInit()
    - Resets the USART peripheral to its default state.

Data Transmission and Reception

- USART_SendData()
    - Sends data using polling mode.
- USART_ReceiveData()
    - Receives data using polling mode.
- USART_SendDataIT()
    - Sends data using interrupt mode.
- USART_ReceiveDataIT()
    - Receives data using interrupt mode.

Interrupt Functions

- USART_IRQInterruptConfig()
    - Enables or disables USART interrupts.
- USART_IRQPriorityConfig()
    - Sets interrupt priority.
- USART_IRQHandling()
    - Handles transmit, receive, idle, CTS, and error interrupts.

Utility Functions

- USART_GetFlagStatus()
    - Checks status flags such as TXE, RXNE, and TC.
- USART_ClearFlag()
    - Clears USART status flags.
- USART_SetBaudRate()
    - Calculates and configures the baud rate register.

Callback Function

- USART_ApplicationEventCallback()
    - Allows the application to handle communication events such as transmission complete, reception complete, idle detection, and communication errors.